Quantifying the sixth-century Arabian drought

Extreme value analysis of the Hoti Cave H12 stalagmite δ^{18}O record

Sandy Hardian Susanto Herho (NIM 32426003)

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Code and data: `https://github.com/sandyherho/utsCuacaIklimEks`

Weather and Climate Extremes
Doctoral Program in Earth Sciences, Bandung Institute of Technology

The claim under test

Fleitmann et al. (2022) combine a new stalagmite record from northern Oman with regional proxies and history.

- Himyar dominated southern Arabia until the Aksumite conquest of 525 CE and dissolved during the sixth century.
- The stalagmite shows its most severe aridity at about 500–530 CE, peaking near 523 ± 30 CE.
- The drought, the authors argue, eroded the resilience of an economy built on rain-fed and runoff agriculture.

Aim

Using only the δ^{18}O series and extreme value statistics, quantify

1. how large the drought was,
2. how rare it was,
3. whether the signal and its rarity are physically and statistically credible.

The archive: stalagmite H12 from Hoti Cave

- Hoti Cave, northern Oman (23°05′N, 57°21′E, 800 m). Annual rainfall 50–255 mm, about two thirds of it in winter and spring from Mediterranean frontal systems (Fleitmann et al., 2022).
- Rain seeps through soil and limestone, reaches the cave as drip water, and deposits calcite layer by layer. The stalagmite grew about 0.23 mm per year.
- Ages come from 20 uranium-thorium dates; the isotope samples have a resolution of about 2 years.

What we can and cannot read

Calcite stores the oxygen isotope ratio of the drip water, which carries the rainfall signal. It does not record rainfall in millimeters directly; the link must be calibrated.

What δ^{18}O measures

Oxygen has a light isotope (^{16}O) and a rare heavy one (^{18}O). With $R = {}^{18}\mathrm{O}/{}^{16}\mathrm{O}$,

$$\delta^{18}\mathrm{O} = \left(\frac{R_{\mathrm{sample}}}{R_{\mathrm{standard}}} - 1 \right) \times 1000 \quad (‰).$$

Here the standard is VPDB. A more negative value means relatively less ^{18}O.

Heavy water molecules condense more readily, so rain is enriched in ^{18}O relative to the vapor it leaves behind (Dansgaard, 1964).

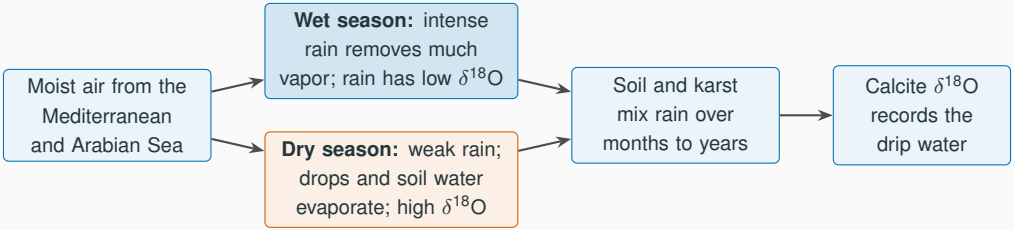
Rayleigh distillation

As a fraction f of the original vapor remains,

$$\delta_{\mathrm{v}} \approx \delta_{\mathrm{v},0} + 1000\,(\alpha - 1)\ln f,$$

with $\alpha \approx 1.0098$ (liquid–vapor, 20 °C). Raining out half of the vapor ($f = 0.5$) lowers the vapor, and the rain that follows, by about 7‰.

The amount effect: why higher δ^{18}O means less rain



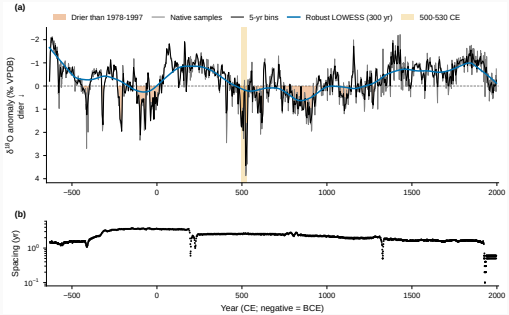
- In the tropics and subtropics, rain δ^{18}O falls as rain amount rises: the amount effect (Dansgaard, 1964; Lachniet, 2009).
- At Hoti it is calibrated: H12 δ^{18}O correlates with instrumental Oman rainfall at $r = -0.69$ (Fleitmann et al., 2022).

At Hoti, dry years push δ^{18}O up through three processes

Process	Mechanism	Dry-year effect
Amount effect	Less rainout; falling drops evaporate	δ^{18}O higher
Soil evaporation	Light rain on dry soil is enriched before it infiltrates	δ^{18}O higher
Cave kinetics	Slow drips degas CO_2 fast; calcite forms out of isotopic equilibrium	δ^{18}O higher

- Kinetic effects at Hoti are confirmed by the strong coupling of δ^{18}O and δ^{13}C ($r^2 = 0.52$) (Fleitmann et al., 2022). Because all three act in the same direction, dry extremes are amplified: δ^{18}O responds nonlinearly to rainfall.
- Temperature is a minor control. Calcite δ^{18}O changes by about -0.2‰ per °C (Lachniet, 2009); a $+3.4$‰ excursion would need more than 15 °C of cave cooling. This study therefore measures how extreme the *proxy* is, not rainfall in millimeters.

The record: one excursion stands out



- **Reading the figure.** (a) Drought plots *downward*. Grey: raw samples; black: 5-yr means; blue: smooth 300-yr trend; orange shading: drier than 1978–1997. The yellow band marks 500–530 CE.
- (b) Time between samples, from 0.1 to 3.8 yr. Raw samples therefore average very different spans of time, which the next step corrects.

Step 1. Put all samples on an even time step

Problem. The samples are evenly spaced along the stalagmite, not in time. A sample that covers 3 years blends a dry year with wetter ones; a sample that covers half a year does not. Extremes would look larger where sampling is fine.

Method. Cut the timeline into 5-yr boxes and average the n_k samples in box k:

$$\bar{x}_k = \frac{1}{n_k} \sum_{i \in k} x_i.$$

Five years exceeds the widest gap (3.8 yr), so every box holds at least one real measurement and nothing is invented.

Result

- 527 evenly spaced values, each representing 5 years.
- Driest value: $+3.43‰$ at 520–525 CE.
- It lies 5.1 standard deviations above the mean of the rest of the record.

Why a special theory for extremes

- Ordinary statistics describe the typical year. A drought study is about the rare years in the tail, where a bell-curve (normal) fit is usually wrong.
- Extreme value theory does for extremes what the central limit theorem does for averages: the largest values follow a small family of curves, whatever the shape of the full data (Coles, 2001).
- It turns “this value is very large” into “a value this large happens, on average, once every T years”.

The idea of block maxima

Hydrologists design dams from the *largest flood of each year*. We do the same with time blocks: keep only the driest value of each block, then study how those driest values are distributed.

Here a year is too short, so the blocks are 50 years (main) and 100 years (check).

Step 2. Cut the record into blocks

Method. Split the 5-yr series into blocks of length B and keep the driest value of each block j:

$$M_j = \max\{\bar{x}_k : k \in \text{block } j\}, \qquad j = 1, \ldots, m.$$

$B = 50$ yr gives $m = 53$ maxima; $B = 100$ yr gives $m = 27$.

Trade-off. Longer blocks hold truer extremes but give fewer of them; shorter blocks give more maxima, some of which are not very extreme.

Independence of the blocks

The theory needs maxima that do not influence each other. Dry and wet spells carry a memory that fades as

$$\rho(\Delta t) = e^{-\Delta t/\tau},$$

with $\tau \approx 11.5$ yr in this record (Mudelsee, 2002). Between blocks 50 yr apart, $\rho \approx 0.01$; at 100 yr, $\rho < 0.001$. Neighboring blocks are practically independent.

Step 3. The generalized extreme value (GEV) distribution

Block maxima follow one curve, the GEV (Coles, 2001). The chance that a block maximum stays *below* a level z is

$$G(z) = \exp\left\{-\left[1 + \xi\,\frac{z-\mu}{\sigma}\right]^{-1/\xi}\right\},$$

valid where $1 + \xi(z-\mu)/\sigma > 0$. When $\xi = 0$ it becomes the Gumbel curve

$$G(z) = \exp\left\{-\exp\left[-\frac{z-\mu}{\sigma}\right]\right\}.$$

The chance of *exceeding* z in one block is $1 - G(z)$.

Three numbers, in plain words

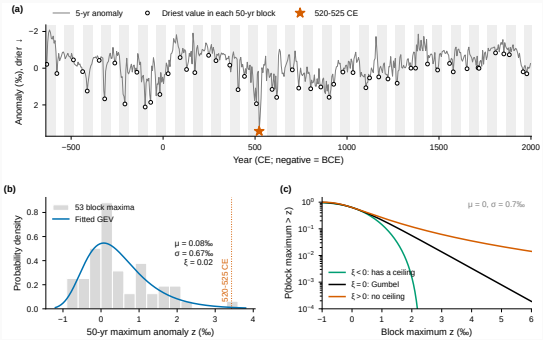
Location μ: the typical driest value of a block.

Scale σ: how much block maxima vary around μ.

Shape ξ: how fast very dry blocks become rare.

- $\xi < 0$: a ceiling exists at $\mu - \sigma/\xi$.
- $\xi = 0$: each extra unit of dryness cuts the chance by the same factor.
- $\xi > 0$: no ceiling.

The block maxima method in one picture



- **Reading the figure.** (a) Grey and white stripes: 50-yr blocks. Circles: the driest value of each block; star: 520–525 CE. (b) Histogram of the 53 block maxima and the fitted GEV curve; the event sits alone at the far right. (c) How the shape ξ changes the chance of a very dry block.

Step 4. Fitting the curve and measuring uncertainty

Maximum likelihood. Choose μ, σ, ξ that make the m observed maxima $z_1, \ldots, z_m$ most probable, by maximizing the log-likelihood (Coles, 2001)

$$\ell(\mu,\sigma,\xi) = -\,m \ln \sigma - \left(1+\tfrac{1}{\xi}\right) \sum_{j=1}^{m} \ln\left[1+\xi\,\tfrac{z_j-\mu}{\sigma}\right] - \sum_{j=1}^{m} \left[1+\xi\,\tfrac{z_j-\mu}{\sigma}\right]^{-1/\xi}.$$

In words: try many curves and keep the one under which our data are least surprising (Python `pyextremes`).

Bootstrap uncertainty

Nature gave us one set of block maxima; a slightly different history would give a slightly different set. The bootstrap imitates this (Efron, 1979):

1. draw m maxima at random, with replacement;
2. refit the GEV;
3. repeat thousands of times.

The spread of the refitted answers is the uncertainty. We report the 5–95 % range.

Step 5. From the curve to return levels and return periods

Return period of a level z, in years:

$$T(z) = \frac{B}{1 - G(z)}.$$

Return level: the dryness reached once every T years,

$$z_T = \mu - \frac{\sigma}{\xi}\left[1 - y_T^{-\xi}\right], \qquad y_T = -\ln\left(1 - \frac{B}{T}\right),$$

and $z_T = \mu - \sigma \ln y_T$ when $\xi = 0$.

Chance of at least one such event in L years:

$$P_L = 1 - \left(1 - \frac{B}{T}\right)^{L/B} \approx 1 - e^{-L/T}.$$

In plain words

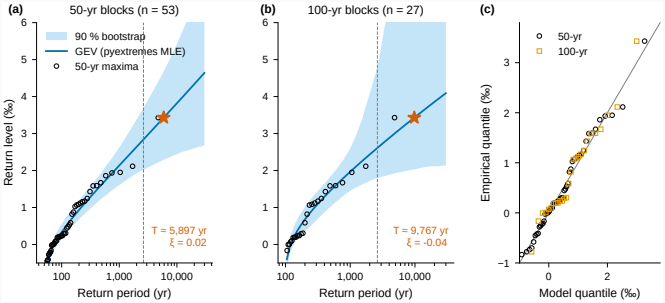
$T = 10{,}000$ yr means a 1-in-10,000 chance in any year, not a schedule.

Checking the fit. Rank the m maxima from smallest ($i = 1$) to largest and give each an observed return period (Gringorten, 1963):

$$p_i = \frac{i - 0.44}{m + 0.12}, \qquad T_i = \frac{B}{1 - p_i}.$$

If the fitted curve passes through these points, the model is adequate.

Result: a return period of several thousand years



- **Reading the figure.** (a, b) How dry a block maximum is for each return period, with 50-yr and 100-yr blocks. Circles: observed maxima (T_i); blue line: fitted GEV; band: 90 % bootstrap range; star: 520–525 CE; dashed line: record length. (c) Observed versus model quantiles: points near the line mean a good fit.
- **Result.** The curve reaches the 520–525 CE value at about 5,897 yr (50-yr blocks) and 9,767 yr (100-yr blocks), well beyond the 2,630-yr record.

Results in numbers

	50-yr blocks	100-yr blocks
Number of maxima m	53	27
Location μ (‰)	0.08	0.42
Scale σ (‰)	0.67	0.72
Shape ξ (5–95 %)	0.02 (-0.21 to 0.16)	-0.04 (-0.52 to 0.68)
1,000-yr return level (‰)	2.14 (1.59–2.65)	1.97 (1.39–2.57)
Return period of 520–525 CE (yr)	5,897	9,767
bootstrap lower bound (5 %)	2,087	1,549
Chance of such an event in 2,630 yr	36 %	24 %

- ξ is close to zero: the tail is nearly Gumbel, with no clear ceiling.
- The 520–525 CE value ($+3.43$‰) lies far above the 1,000-yr return level (about 2‰). Even the bootstrap lower bounds exceed 1,500 yr.

Check: removing the event from its own fit

- The fitted curve includes the 520–525 CE value itself, which pulls the tail toward it and makes the event look less rare.
- Analogy: judging whether a basketball player is unusually tall with an average that already includes that player.
- Test: remove 500–530 CE, refit the GEV, and ask how surprising the event is to the rest of the record.

Result without the event

	50-yr	100-yr
Shape ξ	-0.11	-0.31
T (yr)	54,812	∞

The next driest value is only $+2.12‰$. Without the event, the tail acquires a ceiling: the event becomes about nine times rarer (50-yr blocks) or lies beyond the ceiling (100-yr blocks).

Physical checks: the excursion is a real signal

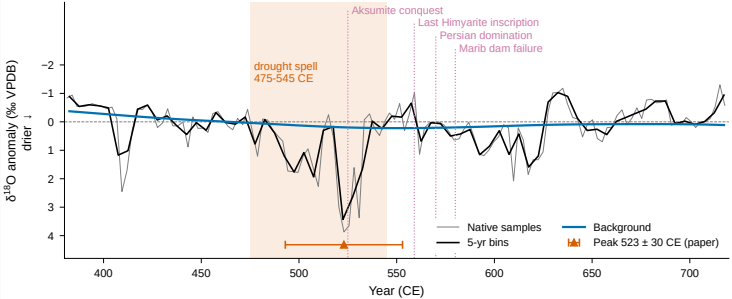
- **Not a single bad analysis.** 4 of the 5 samples above $+2.5$‰ in the whole record lie between 515 and 535 CE, on separate growth layers about 2 mm apart.
- **Not temperature.** At about -0.2‰ per °C, the excursion would require a cave cooling of more than 15 °C (Lachniet, 2009).
- **Independent indicators agree.** During the same interval the paper reports a narrower stalagmite (slower drips) and higher δ^{13}C (less effective moisture) (Fleitmann et al., 2022).
- **Not only a dry century.** The smooth 300-yr trend was just $+0.19$‰ at the time, close to today; the excursion rises far above it.

The excursion is a genuine hydroclimate signal. Its size in ‰ is amplified by the processes on the previous slides.

Physical drivers of the drought proposed by the paper

- **Winter-spring rainfall.** Lake and cave records from Iran, the Levant and Lebanon dry out from about 480–510 CE, while a Turkish lake becomes wetter. This pattern suggests that storm tracks shifted north, so fewer Mediterranean fronts reached Arabia (Fleitmann et al., 2022).
- **Summer monsoon.** Northern Hemisphere summer insolation was near its Holocene minimum; monsoon records from Oman, the Arabian Sea, Ethiopia and India reach their weakest values in the sixth century (Fleitmann et al., 2022).
- **Timing of the Late Antique Little Ice Age.** The cooling began in 536 CE (Büntgen et al., 2016), after the H12 peak, so it cannot have triggered it, although it may have prolonged the stress.
- **Limit of this record.** H12 integrates winter-spring rainfall at one site; it cannot separate these drivers.

Historical context: consistent timing, limited precision



- **Reading the figure.** Drought plots downward; orange box: the 475–545 CE drought spell; orange bar: the peak age and its dating uncertainty; purple lines: historical events.
- The $\pm$30-yr age uncertainty of the peak covers the Aksumite conquest (525 CE) and approaches the last Himyarite inscription (about 559 CE).
- Any event of those decades would match equally well: the timing is consistent with the historical sequence but cannot single it out.

What is robust and what is not

Claim	Evidence	Status
A real excursion, not an error	Several growth layers; temperature cannot explain it	**Strong**
Driest value in 2,600 yr	Rank 1 of 527 values; 5.1 standard deviations	**Strong**
Return period far longer than the record	5,897 and 9,767 yr; bootstrap lower bounds above 1,500 yr	**Strong**
A precise return period	Depends on block size and on whether the event is in the fit	**Weak**
Causal link to Himyar's fall	Timing within $\pm$30 yr; no mechanism in this record	**Weak**

Caveats

- **One archive and one proxy.** A single stalagmite stands in for southern Arabia, and only δ^{18}O is analyzed.
- **Proxy, not rainfall.** The amount effect and cave kinetics make δ^{18}O respond nonlinearly to rainfall, and the event lies far outside the modern calibration range. The return periods describe the proxy, not millimeters of rain.
- **Few maxima.** 53 or 27 block maxima constrain the tail only loosely, which is why the shape ξ and the return period carry wide uncertainty.
- **Ages.** No age-model ensemble is published, so dating uncertainty is discussed but not propagated.
- **Data copy.** The table was transcribed from the NOAA archive; the code reads the original file automatically once it is placed in `data/raw`.

Conclusions

1. The 520–525 CE excursion in H12 is a real hydroclimate signal and the driest value in 2,600 years, 5.1 standard deviations above the rest of the record.
2. The amount effect and cave kinetics both raise δ^{18}O in dry years, so the proxy amplifies the drought.
3. Block maxima with a GEV curve give a return period of about 5,897 to 9,767 years, far longer than the record; removing the event from the fit makes it rarer still.
4. The exact number is uncertain, but the drought was exceptional on a multi-millennial scale. Its link to Himyar rests on timing that dating resolves only to a few decades.

Reproducibility. Code, data, figures and reports (MIT license): `https://github.com/sandyherho/utsCuacaIklimEks`. Running `python scripts/run_all.py` reproduces every number in these slides.

References

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